

# **DATA CLEANING AND PREPROCESSING REPORT**

## **Customer Personality Analysis Dataset**

### **Submitted By**

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### **Tools Used**

- Python
  - Pandas
  - VS Code
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## **1. Objective**

The objective of this project is to clean and preprocess a raw dataset by identifying and handling missing values, removing duplicate records, standardizing data formats, validating data types, and preparing the dataset for further analysis.

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## **2. Dataset Overview**

Dataset Name: Customer Personality Analysis

Source: Kaggle

The dataset contains customer demographic information, purchasing behavior, and campaign response details.

Key Features:

- ID
- Year\_Birth
- Education
- Marital\_Status
- Income
- Kidhome
- Teenhome
- Dt\_Customer
- Recency
- Product Spending Information

Initial Dataset Shape:

- Rows: 2240

- Columns: 29

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### 3. Data Quality Issues Identified

During the initial inspection, the following issues were found:

#### Missing Values

The Income column contained missing values.

#### Duplicate Records

The dataset was checked for duplicate entries.

#### Data Type Validation

Numerical and categorical columns were verified to ensure correct data types.

#### Date Formatting

The Dt\_Customer column required conversion to datetime format.

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### 4. Data Cleaning Process

#### Step 1: Load Dataset

The dataset was loaded using Pandas.

```
import pandas as pd
```

```
df = pd.read_csv("raw_data.csv", sep="\t")
```

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#### Step 2: Missing Value Analysis

Missing values were identified using:

```
df.isnull().sum()
```

#### Screenshot 1:

Insert Missing Values Before Cleaning Screenshot

Observation:

The Income column contained missing values that required treatment.

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#### Step 3: Missing Value Treatment

Missing numerical values were replaced using the median.

Reason:

Median is less affected by outliers and provides a reliable representation of the data distribution.

```
df["Income"] = df["Income"].fillna(df["Income"].median())
```

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#### **Step 4: Duplicate Record Removal**

Duplicate records were identified and removed.

```
df.drop_duplicates()
```

#### **Screenshot 2:**

Insert Duplicate Records Screenshot

Observation:

Duplicate entries were removed to improve data quality and avoid biased analysis.

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#### **Step 5: Column Name Standardization**

Column names were standardized by:

- Converting to lowercase
- Removing extra spaces
- Replacing spaces with underscores

Example:

- Year\_Birth → year\_birth
  - Marital\_Status → marital\_status
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#### **Step 6: Date Conversion**

The Dt\_Customer column was converted into datetime format.

```
df["dt_customer"] = pd.to_datetime(  
    df["dt_customer"],  
    dayfirst=True,  
    errors="coerce"  
)
```

#### **Screenshot 3:**

Insert Data Type Verification Screenshot

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## Step 7: Final Validation

The dataset was checked again to ensure:

- No missing values remained
- Data types were correct
- Data was ready for analysis

```
df.isnull().sum()
```

## Screenshot 4:

Insert Missing Values After Cleaning Screenshot

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## 5. Results

The following improvements were achieved:

- ✓ Missing values handled successfully
  - ✓ Duplicate records removed
  - ✓ Column names standardized
  - ✓ Date column converted into datetime format
  - ✓ Data types validated
  - ✓ Dataset prepared for analysis
  - ✓ Clean dataset exported successfully
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## 6. Output

The cleaned dataset was exported as:

```
cleaned_data.csv
```

## Screenshot 5:

Insert Cleaned Dataset Preview Screenshot

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## 7. Conclusion

The Customer Personality Analysis dataset was successfully cleaned and preprocessed using Python and Pandas. Data quality issues such as missing values, duplicate records, and formatting inconsistencies were identified and resolved. The resulting dataset is clean, consistent, and ready for further analysis, visualization, and machine learning applications.